

Initial Project Planning Template

Date	July 9, 2026
Team ID	SWTID-2026-1262
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create a product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Epic 1: Environment Setup	USN-1	As a developer, I can configure a local Python 3.13 / Anaconda virtual workspace to isolate dependencies cleanly.	2	High	Hari Charan	July 09, 2026	July 12, 2026
Sprint-1	Epic 2: Data Preprocessing	USN-2	As a developer, I can synthetically generate a balanced 1500-row agricultural feature matrix to represent diverse ecological boundaries.	3	High	Hari Charan	July 09, 2026	July 12, 2026
Sprint-2	Epic 3: Model Training	USN-3	As a data scientist, I can train multiple algorithms (KNN, Logistic Regression, Decision Tree, Random Forest) over split training matrices.	5	High	Hari Charan	July 09, 2026	July 12, 2026
Sprint-2	Epic 4: Model Persistence	USN-4	As a backend system, I can serialize the trained Random Forest classifier into a persistent .pkl file format for reuse.	1	High	Hari Charan	July 09, 2026	July 12, 2026

Sprint-3	Epic 5: Flask Backend Core	USN-5	As a system web server, I can safely deserialize and load model.pkl into RAM at system startup.	2	High	Hari Charan	July 09, 2026	July 12, 2026
Sprint-4	Epic 6: UI Design	USN-6	As a web user, I can view an interactive tabbed dashboard containing input forms styled cleanly using responsive Bootstrap 5.	3	High	Hari Charan	July 09, 2026	July 12, 2026
Sprint-5	Epic 7: Scenario 1 Engine	USN-7	As a farmer, I can input soil measurements to instantly receive an AI-powered crop recommendation optimization output.	3	High	Hari Charan	July 09, 2026	July 12, 2026
Sprint-5	Epic 8: Scenario 2 Engine	USN-8	As a researcher, I can cross-validate a chosen target crop against calculated environmental metrics to catch compatibility variances.	3	Medium	Hari Charan	July 09, 2026	July 12, 2026
Sprint-6	Epic 9: Scenario 3 Engine	USN-9	As a policymaker, I can review specialized regional data profiles to extract agro-ecological zoning limits and sustainability maps.	3	Medium	Hari Charan	July 09, 2026	July 12, 2026
Sprint-6	System Testing	USN-10	As a tester, I can perform local validation loops using the native Flask development console to ensure zero runtime crashes.	2	High	Hari Charan	July 09, 2026	July 12, 2026